

A full construction of FUNTFs in real ℓ_1^n

Agent lane 18 packet for run fa_banach_001

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Status

This is a candidate full solution, subject to human review, of Question 8.2 in Chavez-Dominguez, Freeman, and Kornelson [1]. The question asks whether real ℓ_1^n has a length N finite unit norm tight frame (FUNTF) for every $N \geq n$. The construction below gives such a frame for every pair of integers $n \geq 1$ and $N \geq n$.

Source question evidence

The open problem appears on page 14, Section 8, Questions 8.1 and 8.2 of [1]. The relevant source crop is included below.

□

8. OPEN PROBLEMS

We showed in Section 4 that a large class of finite dimensional Banach spaces have finite unit norm tight frames of every length at least the dimension of the space. However, we do not have any examples where this is not possible.

Question 8.1: Does every n dimensional Banach space have a length N FUNTF for all $N \geq n$?

It seems very difficult to create a method of constructing FUNTFs of arbitrary size that works in any finite dimensional Banach space. Thus, it may be best to focus first on specific classical Banach spaces. We have shown in Section 4 that complex ℓ_1^n has a FUNTF of length N for all $n \leq N$ and in Section 7 that real ℓ_1^2 , ℓ_1^3 , and ℓ_1^4 each have FUNTFs of all lengths at least their dimension.

Question 8.2: Does real ℓ_1^n have a length N FUNTF for all $N \geq n$?

Note that for each n , there are only finitely many values of N for which we do not know the answer to the question above. Indeed, it follows from our results that for any $N \geq n(n-1)$ a FUNTF of length N does exist: if $N = n(n-1+m) + k$ with $0 \leq k \leq n-1$ and $m \geq 0$, we can take the union of k copies of a FUNTF of length $n+1$ and $n-1-k+m$ copies of a FUNTF of length n .

A finite dimensional Banach space X has a length N FUNTF if and only if a scalar multiple of the identity operator on X can be expressed as a sum of N normalized rank 1 projections. In the Hilbert space case, operators which can be written as sums of norm one rank one projections are characterized as positive operators with integer trace [KL04]. Operators on complex Banach

Question 8.2 asks: “Does real ℓ_1^n have a length N FUNTF for all $N \geq n$?” The same page notes that the source paper already constructs lengths n and $n + 1$, so for fixed n it remains only to construct the finite gap of lengths $n + 2, \dots, 2n - 1$.

Definitions

For a finite-dimensional Banach space X , the source paper calls a sequence $(x_j, x_j^*)_{j=1}^N \subset X \times X^*$ a finite unit norm tight frame if

$$\|x_j\| = \|x_j^*\| = x_j^*(x_j) = 1 \quad (1 \leq j \leq N),$$

and its frame operator

$$S(x) = \sum_{j=1}^N x_j^*(x) x_j$$

is a scalar multiple of the identity. Taking traces shows that in dimension n the scalar must be N/n .

Novelty check

Before promotion, the local run indexes were searched for

1804.03677, real ell_1^n, FUNTF, finite unit norm tight frame.

The only relevant local result was the earlier Hadamard-order partial packet from this same run. Web searches on 2026-06-26 for exact phrases around real ℓ_1^n FUNTFs, finite unit norm tight frames in Banach ℓ_1 , and the source title surfaced the source arXiv record but no direct later answer. This novelty check is bounded and should be reviewed by a human.

Proof Intuition

The source paper observes that a length N FUNTF in ℓ_1^n is the same as a decomposition of $(N/n)I$ into N norm-one rank-one projections. In real ℓ_1^n , sign vectors are natural norming functionals. The trick is to take n sign vectors with exactly one positive coordinate, add $k = N - n$ all-positive sign vectors, and choose only two coordinate weights α and β . The diagonal terms are forced by normalization. The off-diagonal terms then all have the same value by symmetry, and the single linear equation

$$-\alpha + (n - 3)\beta + \frac{k}{n} = 0$$

chooses the weights. This fills every finite-gap length $n + k$ with $0 \leq k \leq n - 1$. Unions of tight frames then give all larger lengths.

Theorem 1. *For every pair of integers $n \geq 1$ and $N \geq n$, real ℓ_1^n has a length N FUNTF.*

Proof. Let $(e_i)_{i=1}^n$ be the standard basis of ℓ_1^n , with biorthogonal coordinate functionals $(e_i^*)_{i=1}^n$.

We first handle $n \geq 3$ and the finite-gap lengths $N = n + k$ with $0 \leq k \leq n - 1$. Put

$$\beta = \frac{n - k}{2n(n - 2)}, \quad \alpha = 1 - (n - 1)\beta.$$

Since $0 \leq k \leq n-1$ and $n \geq 3$, both α and β are nonnegative, and $\alpha + (n-1)\beta = 1$.

For $1 \leq j \leq n$, define

$$x_j = \alpha e_j - \beta \sum_{i \neq j} e_i, \quad x_j^* = e_j^* - \sum_{i \neq j} e_i^*.$$

For $1 \leq t \leq k$, define

$$y_t = \frac{1}{n} \sum_{i=1}^n e_i, \quad y_t^* = \sum_{i=1}^n e_i^*.$$

The x_j vectors satisfy

$$\|x_j\|_1 = \alpha + (n-1)\beta = 1, \quad \|x_j^*\|_\infty = 1, \quad x_j^*(x_j) = \alpha + (n-1)\beta = 1.$$

Similarly, $\|y_t\|_1 = \|y_t^*\|_\infty = y_t^*(y_t) = 1$.

Let S be the frame operator of these $n+k$ pairs. Fix m . The coefficient of e_m in $S(e_m)$ is

$$\alpha + (n-1)\beta + \frac{k}{n} = 1 + \frac{k}{n} = \frac{n+k}{n}.$$

Now fix $\ell \neq m$. The coefficient of e_ℓ in $S(e_m)$ is the sum of: the contribution $-\beta$ from $j = m$, the contribution $-\alpha$ from $j = \ell$, the contribution $+\beta$ from each of the remaining $n-2$ indices, and the contribution $1/n$ from each of the k all-positive pairs. Hence it equals

$$-\beta - \alpha + (n-2)\beta + \frac{k}{n} = -\alpha + (n-3)\beta + \frac{k}{n}.$$

Using $\alpha = 1 - (n-1)\beta$ and the definition of β , this is

$$-1 + 2(n-2)\beta + \frac{k}{n} = -1 + \frac{n-k}{n} + \frac{k}{n} = 0.$$

Therefore $S(e_m) = ((n+k)/n)e_m$ for every m , so $S = ((n+k)/n)I_{\ell_1^n}$. Thus real ℓ_1^n has a length $n+k$ FUNTF for every $0 \leq k \leq n-1$ when $n \geq 3$.

The case $n = 1$ is immediate: take N copies of (e_1, e_1^*) .

For $n = 2$, there is a length 2 FUNTF, namely the standard Auerbach basis $(e_1, e_1^*), (e_2, e_2^*)$. There is also a length 3 FUNTF:

$$(e_1, e_1^*), \quad \left(\frac{1}{4}e_1 + \frac{3}{4}e_2, e_1^* + e_2^*\right), \quad \left(-\frac{1}{4}e_1 + \frac{3}{4}e_2, -e_1^* + e_2^*\right).$$

Each pair is normalized and norming, and its frame operator is

$$e_1^* \otimes e_1 + (e_1^* + e_2^*) \otimes \left(\frac{1}{4}e_1 + \frac{3}{4}e_2\right) + (-e_1^* + e_2^*) \otimes \left(-\frac{1}{4}e_1 + \frac{3}{4}e_2\right) = \frac{3}{2}I_{\ell_1^2}.$$

Hence ℓ_1^2 has seed FUNTFs of lengths 2 and 3.

Finally, we use unions. If F is a length L FUNTF in ℓ_1^n and G is a length M FUNTF in the same space, then their concatenation is a length $L+M$ FUNTF, since its frame operator is

$$\frac{L}{n}I + \frac{M}{n}I = \frac{L+M}{n}I.$$

For $n \geq 3$ and arbitrary $N \geq n$, write $N = qn + r$ with $0 \leq r < n$. If $r = 0$, use q copies of the length n construction above. If $1 \leq r < n$, use $q-1$ copies of the length n construction and one copy of the length $n+r$ construction. For $n = 2$, every even N is a union of length 2 seeds and every odd $N \geq 3$ is one length 3 seed plus length 2 seeds. This proves the theorem for all $n \geq 1$ and $N \geq n$. $\square$

Scope and relation to the source paper

Theorem 1 gives a positive answer to Question 8.2 as stated. It does not answer the broader Question 8.1 asking whether every n -dimensional Banach space has FUNTFs of every length $N \geq n$, and it does not characterize all sums of norm-one rank-one projections.

Verification notes

The included script

```
code/verify_real_l1_funtf_all_lengths.py
```

constructs the frames described above and checks, in exact rational arithmetic, the norming identities and the frame-operator identity for $1 \leq n \leq 10$ and $n \leq N \leq 4n + 7$. From the packet directory, the command used was:

```
conda run --no-capture-output -n sandbox python  
code/verify_real_l1_funtf_all_lengths.py
```

This computation is not a proof; it is a sanity check for the formula and the small-dimensional edge cases.

Human-review recommendation

Review should focus on the matrix-entry convention in the frame operator and on the finite-gap construction for $n \geq 3$. The proof is otherwise an explicit calculation plus the standard union observation.

References

- [1] J. A. Chavez-Dominguez, D. Freeman, and K. Kornelson, *Frame potential for finite-dimensional Banach spaces*, arXiv:1804.03677.